SIMATS Engineering • CSE • CSA0802 — Python Programming • 24-07-2026 • Category: Analytical • Student Submissions Report

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Q1. Harshad number

ANSWER

```
n = int(input())

s = 0
t = n

while t > 0:
    s += t % 10
    t //= 10

if n % s == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")

print("Harshad numbers from 1 to 500:")

for i in range(1, 501):
    s = 0
    t = i
    while t > 0:
        s += t % 10
        t //= 10
    if i % s == 0:
        print(i, end=" ")
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))

while b ≠ 0:
    temp = b
    b = a % b
    a = temp

print("GCD is", a)
```

INPUT

```
10
20
```

OUTPUT

```
Enter first: Enter second: GCD is 10
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen = set()
    pairs = []

    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
        seen.add(num)

    return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))

result = find_pairs(data, t)

print("Pairs:", result)
```

Q4. Bitwise Operators

ANSWER

Output:
a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32
a - integer
b- integer
c- integer
result-integer

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low ≤ high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        if arr[low] ≤ arr[mid]:
            if target ≥ arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if target > arr[mid] and target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))

result = rotated_binary_search(data, t)
print("Found at:", result)
```

INPUT

```
5
```

OUTPUT

```
Target: Found at: 1
```